



United International University

Department of Computer Science and Engineering

CSE 4883: Digital Image Processing Mid: Fall 2025

Total Marks: 20 Time: 1 hour and 30 minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all the questions. The numbers on the right of the questions denote their marks.

1. A thermal camera measures **small temperature variations** on the human face for fever screening. Which parameter is more important for reliable temperature discrimination—spatial resolution or intensity quantization? Justify your answer. (2)
2. Suppose you have charted the performance of your model in the following confusion matrix based on Email classification for a mailing system: (4)

Predicted Category	Actual Email Category			
	Spam	Promotional	Social	Important
Spam	820	70	60	50
Promotional	60	800	90	50
Social	70	90	780	60
Important	50	40	70	840

Table 1: Confusion matrix showing email categories

- (a) Calculate the **Accuracy, Class-wise Precision, Recall and F1 Score**.
 - (b) Calculate the Macro and Weighted F1 scores.
 - (c) Show the ROC curve by calculating the ratio.
3. After analyzing an **X-ray image(grayscale)**, you came up with the following color intensity distribution: (5)

Color Intensity	Appearance in the Image
135	30
225	12
210	15
20	25
55	18
185	44

Table 2: Pixel Intensity distribution of an image

- (a) Use “**Huffman Coding**” technique to reduce the coding redundancy of this image. Draw the **Huffman tree** and build the **Huffman table** showing the memory requirement.
 - (b) Find the compressed Image size. Also, find the **Compression Ratio, Relative Data Redundancy, and Bits per Pixel**.
 - (c) Comment whether the compression is **lossy or lossless**.
4. Consider the following ANN: (The Input, Output, and weight matrices for different layers are also given) (5)

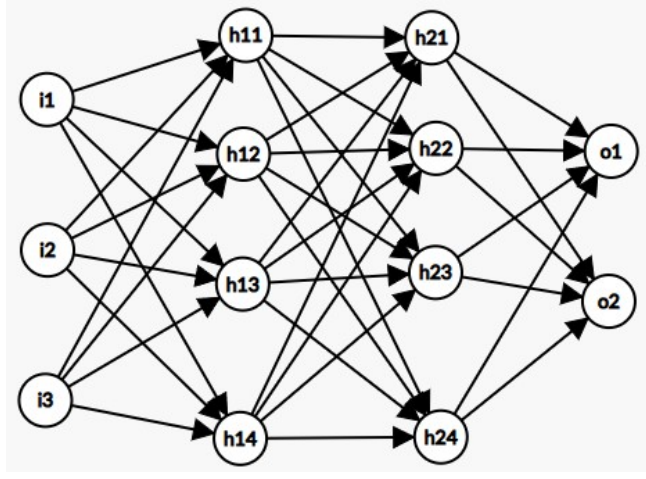


Figure 1: A sample ANN

$$\text{Inputs} = [0.35 \quad 0.20 \quad 0.7], \quad \text{Outputs} = [1 \quad 0]$$

$$W_{ih1} = \begin{bmatrix} 0.15 & 0.30 & 0.20 & 0.35 \\ 0.25 & 0.40 & 0.10 & 0.25 \\ 0.20 & 0.25 & 0.4 & 0.15 \end{bmatrix}$$

$$W_{h1h2} = \begin{bmatrix} 0.1 & 0.30 & 0.25 & 0.35 \\ 0.35 & 0.10 & 0.25 & 0.30 \\ 0.1 & 0.2 & 0.55 & 0.15 \\ 0.25 & 0.05 & 0.35 & 0.35 \end{bmatrix}$$

$$W_{h2o} = \begin{bmatrix} 0.35 & 0.65 \\ 0.55 & 0.45 \\ 0.2 & 0.8 \\ 0.75 & 0.25 \end{bmatrix}$$

Activation functions for the layers: **ReLU(layer 1)**, **Sigmoid(layer 2)**, and suitable(layer 3/output layer) based on the output.

- Now, for the given weights and input, determine the output of the ANN.
- Calculate the loss using log loss function given as follows:

$$\text{Log Loss} = -y_i \log(o_{outi}) - (1 - y_i) \log(1 - o_{outi})$$

5. Consider the following **5x5 grayscale image** where **V= [0,127]**

(4)

132	125	105	100	85
103	185	95	65	70
210	114	72	98	110
155	25	2225	100	62
55	125	45	145	20

Table 3: Pixels of an image

- Find out and show the connectivity of all the 4-adjacent pixels.
- Find out and show the connectivity of all the 8-adjacent pixels.
- Find out and show the connectivity of all the m-adjacent pixels.